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A. Details on Other Attempts

k	Cross-Play	Self-Play
1	1.06 ± 0.04	1.04 ± 0.06
2	0.95 ± 0.18	0.99 ± 0.33
3	1.49 ± 0.11	2.63 ± 0.34
4	2.48 ± 0.22	5.89 ± 0.75
5	2.04 ± 0.35	7.22 ± 0.56

Table 2. Cognitive Hierarchies Performance. We train CH for 5 levels with 3 seeds. The cross-play and self-play results are computed by averaging scores of intra-level pairing of agents trained with different seeds. Cross-play score is averaged over 6 pairs and self-play score is averaged over 3 pairs for each cell.

In this section we provide more details on other attempts mentioned previously.

The core idea behind cognitive hierarchies (CH) (Stahl, 1993) and k-level reasoning (Costa-Gomes et al., 2001) is to train a sequence of K agents of different capabilities. The final agents may hopefully learn strategies that cross-play well through such explicit route of evolution. In our implementation of CH, the first agent $a^{(0)}$ in the sequence is a random agent that pick actions uniformly regardless of the state. The kth agent $a^{(k)}$ is trained to be the "best response" to the pool of agents $\{a^{(0)}, ..., a^{(k-1)}\}$. Intuitively, this means that the first agent will learn to play based only on the hinted facts, i.e. ground information, to play Hanabi because $a^{(0)}$'s actions contain no intentions nor conventions. $a^{(2)}$ can then learn to give more useful hints and the subsequent agents may learn more complicated behaviors. In k-level, the kth agent $a^{(k)}$ only learns to best respond $a^{(k-1)}$ and other aspects remain the same. Because the agents are trained with a random agent $a^{(0)}$ and they will "bomb out" inevitably, we alter the reward scheme so that the agents receive reward 0, instead of the negative of current score, when they lose all life tokens.

The performance of CH is shown in Table 2. The most prominent phenomenon is that CH converges quite slowly, due to the fact that it needs to cooperate with a pool of different yet primitive policies. For each level, we train the models until convergence and 5 levels normally take several days to complete. For reference, it roughly takes less than 20 hours for SAD and our other-play agents to reach 23 points in self-play under the same settings and hardware. The prohibitive cost in time and computation make CH unsuitable for complicated tasks like MARL in Hanabi. Moreover, even though the self-play score is low, we can already see a clear performance gap between self-play and cross-play, making it safe to assume that CH will not work well in zero-shot coordination.

In Table 3, we show results of K-Level method trained for 10 levels. Despite the gap between cross-play and self-

k	Cross-Play	Self-Play
1	0.73 ± 0.18	0.95 ± 0.36
2	0.50 ± 0.02	0.54 ± 0.13
3	2.99 ± 0.11	3.24 ± 0.24
4	1.71 ± 0.12	2.57 ± 0.11
5	6.14 ± 0.58	7.27 ± 1.48
6	2.08 ± 0.22	4.52 ± 1.05
7	6.28 ± 0.92	8.82 ± 2.17
8	1.82 ± 0.25	4.89 ± 1.95
9	6.87 ± 0.91	10.26 ± 2.52
10	2.05 ± 0.28	6.54 ± 2.65

Table 3. K-Level Performance. We train K-Level with $K = 10$ and 3 seeds. The cross-play and self-play scores are computed by intra-level pairing of agents trained with different seeds.

play being smaller, this method suffers from non-monotonic improvements between levels and the same high cost in time and sample complexity.

	Population1	Population2
Population1	23.46 ± 0.01	19.97 ± 0.06
Population2	20.16 ± 0.06	23.44 ± 0.01

Table 4. Performance of Population Based Method. Each cell is computed by pairing all agents from one population with those from the other population and then average the scores. Diagonal can be seen as self-play score and non-diagonal can be seen as cross-play score under a population setting.

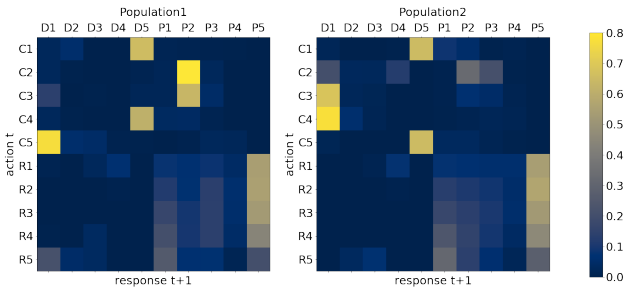

Figure 7. $P(a_t^i \mid a_{t-1}^j)$ matrices of one model from each population. The semantic of the visualization is identical to Figure 5.

Different from CH and K-Level where agents are trained sequentially, population based approaches (Fitzgerald, 2019; Tieleman et al., 2018; Lowe et al., 2019) train agents simultaneously by pairing distinct agents together to generate samples. We briefly experimented with a simple population setting where we initialize N different agents with their private replay buffers. They are uniformly paired together with each other to generate samples and write their observation action sequences into the their own buffer. Each agent is optimized independently at each training step. We train 2 populations with different seeds. Each of them contains 4 agents initialized differently. The numerical results are shown in Table 4. This method can achieve decent cross-play scores. It is worth noting that the diversity between the hyper-parameters of the two populations is much smaller than that of the experiments shown in Figure 4 so that they are not directly comparable. However, a closer look at their respective policy through $P(a_t^i \mid a_{t-1}^j)$ matrices reveals the problem. The way they use color hints not only differs greatly from each other but also breaks the color symmetry of the game, which is the exact problem other-play tries to solve. Qualitatively they are hard for human to play with. They manage to achieve good cross-play scores because the agents seldom use color hints.